

UAV Communications (EE798T) - Assignment 1

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Question 1: Weather Condition Simulation

The simulation was carried out in the MAVproxy terminal in the Ubuntu distribution of the Linux operating system.

a. Procedure for Simulation:

- Launch SITL (Software in the loop) simulation of Arducopter quad using the command "**sim_vehicle.py -v copter --console --map -L iitk**"
- After launching SITL, the mission is planned using the options available in the SITL console, and the waypoints were edited accordingly using the **Waypoint Editor** application from the console.
- Now the flight mode is changed to guided to arm the motors and manually take off to mission altitude using command "**mode guided**".
- The motors are armed using the command "**arm throttle**" and the vehicle is commanded to take off to an altitude of 30m using the command "**takeoff 30**".
- As soon as the vehicle reaches the altitude of 30m, the mode is changed to auto to make the vehicle follow the uploaded waypoints. This is done using the command "**mode auto**".
- Now for simulating the weather conditions, the command "**param set SIM_WIND_SPD 8**" (Roll no: 251010068) to control wind speed and the command "**param set SIM_WIND_DIR <direction>**" were used.
- The parameters were changed intermittently during each part of the simulation to arrive at the required final simulation.

b. Roll and Pitch plots vs Time

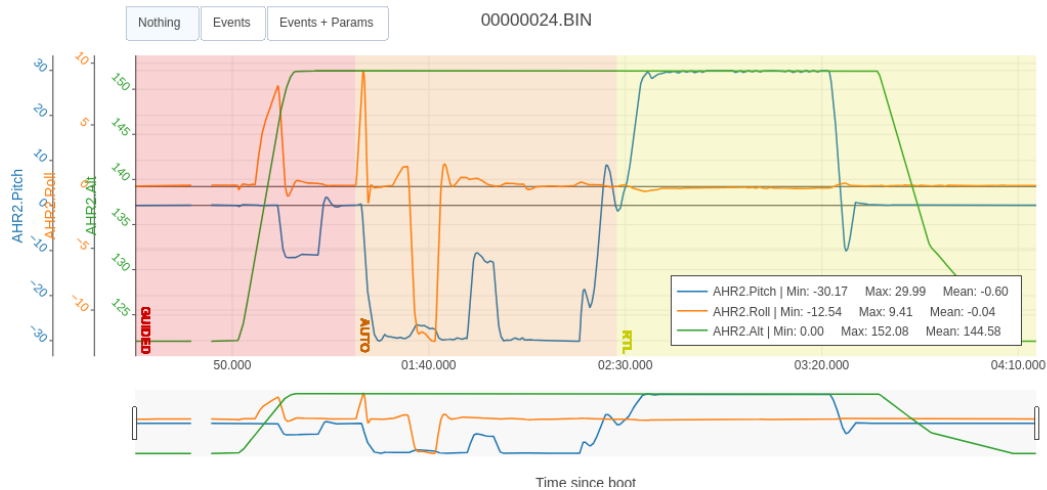


Figure 1: Complete Mission (Altitude, Pitch, Roll vs Time)

For analysis, each part of the mission was manually marked and plotted.

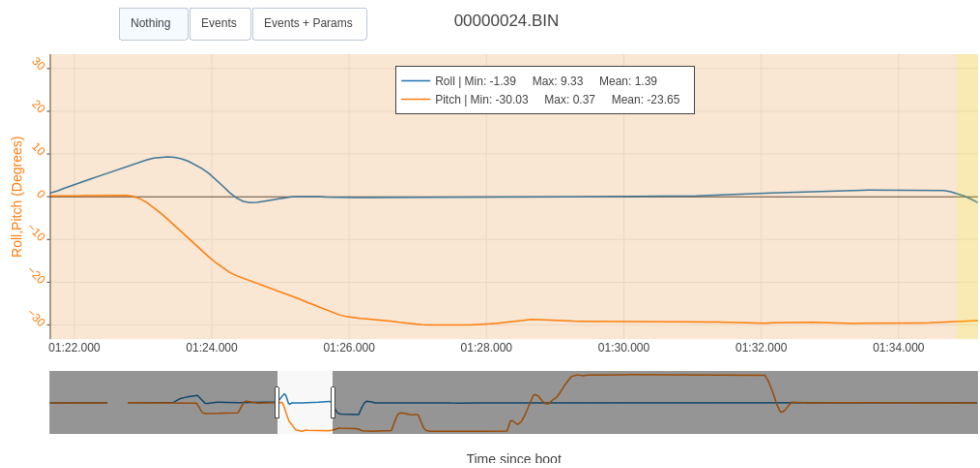


Figure 2: Launch to Waypoint 1 (Pitch, Roll vs Time)

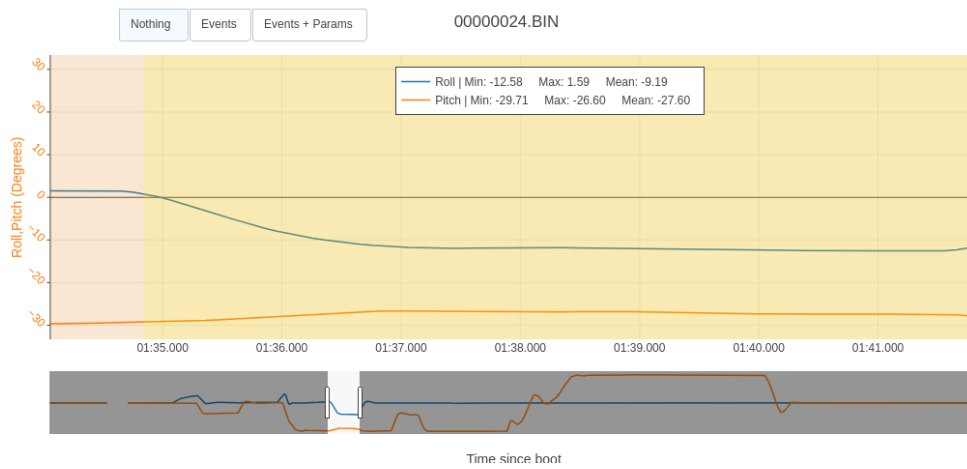


Figure 3: Waypoint 1 to Waypoint 2 (Pitch, Roll vs Time)

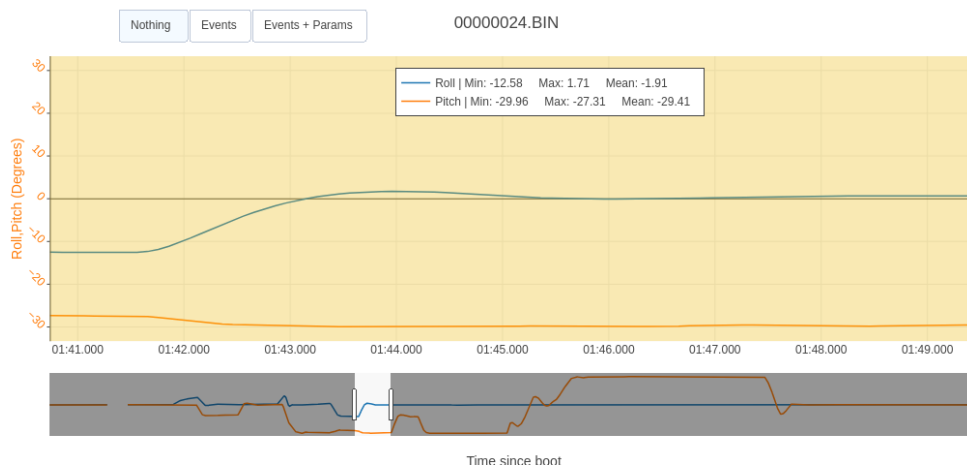


Figure 4: Waypoint 2 to Waypoint 3 (Pitch, Roll vs Time)

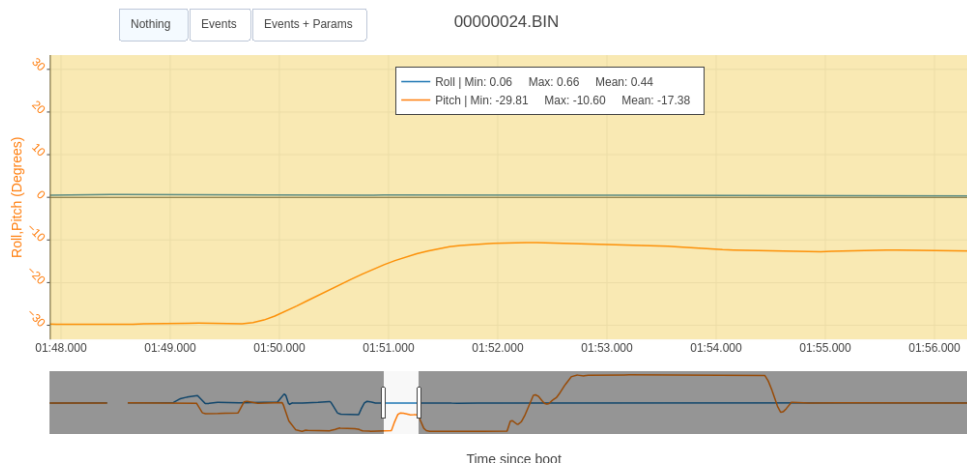


Figure 5: Waypoint 3 to Waypoint 4 (Pitch, Roll vs Time)

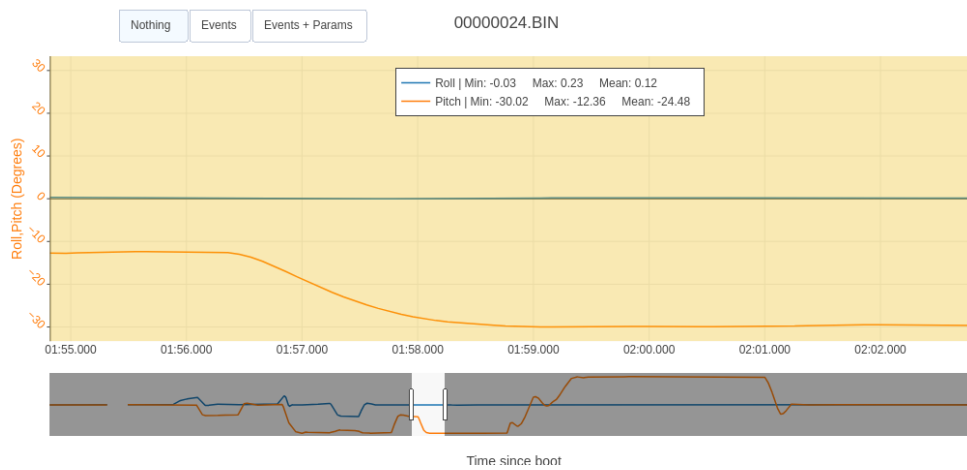


Figure 6: Waypoint 4 to Waypoint 5 (Pitch, Roll vs Time)

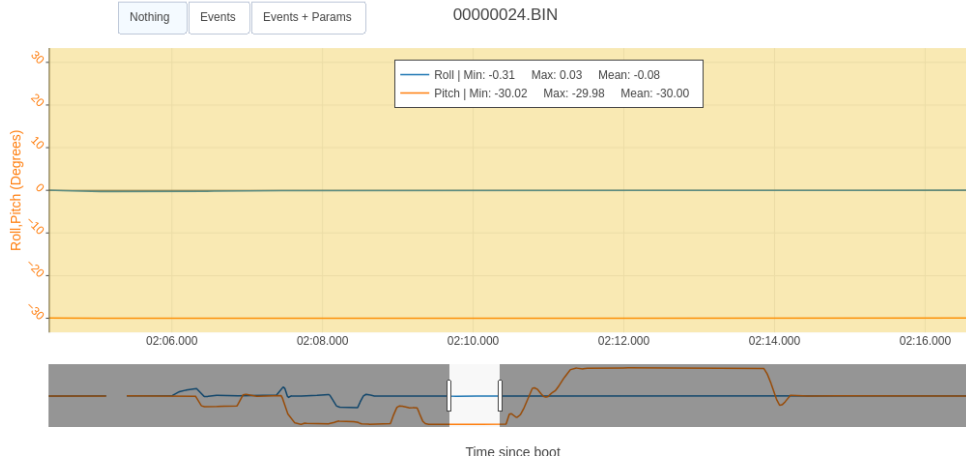


Figure 7: Waypoint 5 to Waypoint 6 (Pitch, Roll vs Time)

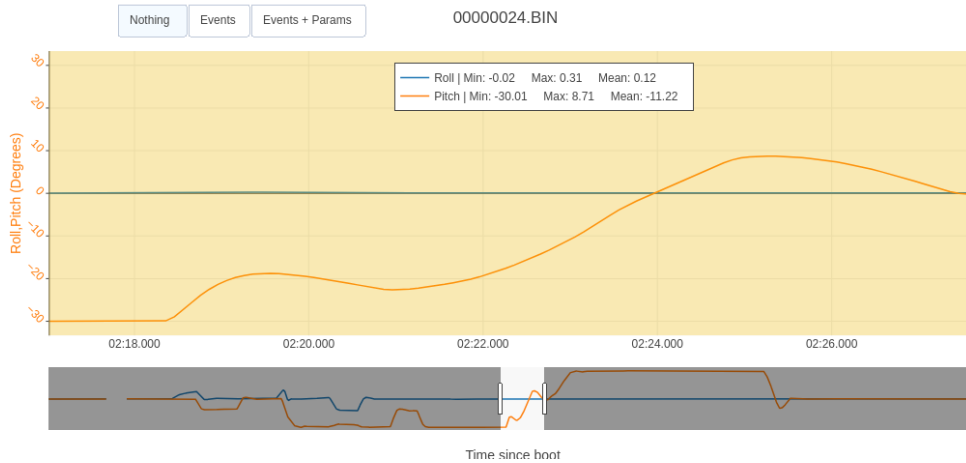


Figure 8: Waypoint 6 to Waypoint 7 (Pitch, Roll vs Time)

c. Explanation

Throughout the mission the vehicle has a bearing of 90° (vehicle starts facing north and yaws right by an angle of 90 degrees facing east).

- It can be observed from fig. 2 that the mean roll angle is 0° and mean pitch angle is about -23.65° (pitching down). During this part of the mission there is zero wind.
- It can be observed from fig. 3 that the mean roll angle is -9.19° roll towards left and mean pitch angle is about -27.60° (pitching down). During this part of the mission there is a crosswind from north to south and the vehicle is heading with a bearing of 90 degrees.
- It can be observed from fig. 4 that the mean roll angle is approx. 0° and mean pitch angle is about -29.41° (pitching down). During this part of the mission there is zero wind.

- It can be observed from fig. 5 that the mean roll angle is approx. 0° roll towards left and mean pitch angle is about -17.38° (pitching down). During this part of the mission there is a tailwind from west to east. And hence to counter the tailwind the drone reduced its pitch angle.
- It can be observed from fig. 6 that the mean roll angle is approx. 0° and mean pitch angle is about -24.48° (pitching down). During this part of the mission there is zero wind.
- It can be observed from fig. 7 that the mean roll angle is approx. 0° roll towards left and mean pitch angle is about -30° (pitching down). During this part of the mission there is a headwind from east to west. And hence to counter the headwind the vehicle reaches the highest average pitch angle of the mission.
- It can be observed from fig. 8 that the mean roll angle is approx. 0° and mean pitch angle is about -11.22° (pitching down). During this part of the mission there is zero wind and vehicle reaches a state of hover at the end.

d. Power Consumption

For calculation of power during each mission segment the graphs (fig. 8 - fig. 12) for battery currents were plotted and peak current during each mission segment was used to calculate power consumption. The table below summarizes the power consumption of the vehicle. The battery by default in Ardupilot is 3s (4.2*3 = 12.6V) is used for power calculation.

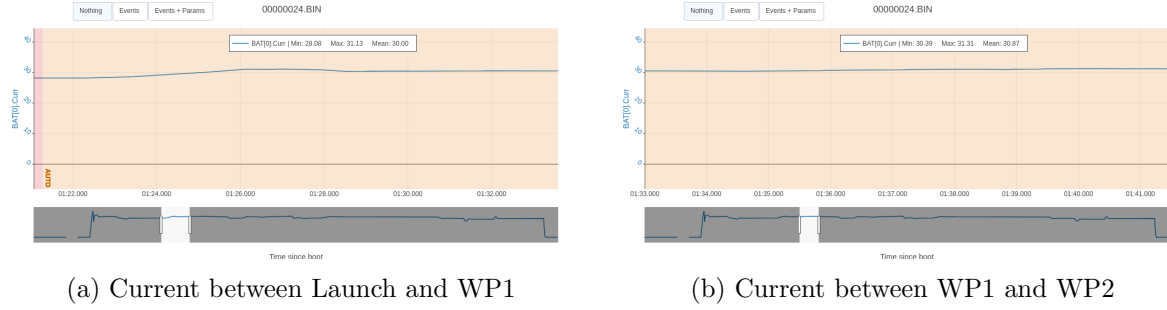


Figure 9: Current Analysis Part 1

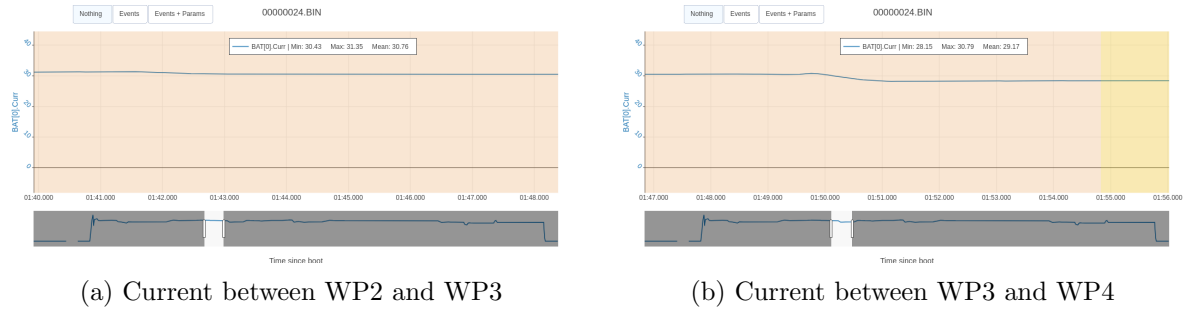


Figure 10: Current Analysis Part 2

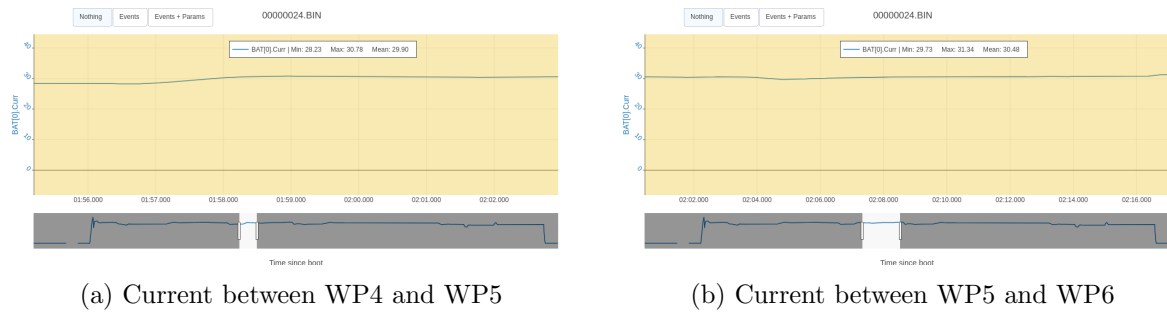


Figure 11: Current Analysis Part 3



Figure 12: Current (Ampere) between Waypoint 6 and Waypoint 7

Table 1: Power Consumption Summary across Mission Segments

Waypoints	Peak Current (A)	Power Consumption (W)	Remarks
Launch – WP1	31.13	392.238	Nominal cruise
WP1 – WP2	31.31	394.506	Crosswind – Power consumed is higher than nominal case
WP2 – WP3	31.35	395.01	Nominal cruise
WP3 – WP4	30.79	387.954	Tailwind – Power consumed is lower than nominal case
WP4 – WP5	30.78	387.828	Nominal cruise
WP5 – WP6	31.35	395.01	Headwind – High power consumption
WP6 – WP7	31.23	393.498	Nominal cruise

However, there are some high values of power consumed for nominal cruise case too. These might be due to the discrepancies in manually commanding wind or manually time-stamping way points.

e. Speed Analysis

The groundspeed of the vehicle is recorded and plotted in fig. 13. I



Figure 13: Ground speed during complete mission

The most vivid observation one can make by looking at the figure one can say that there is a drop in the speed around (2:10) and a small peak around (1:50). The peak and drop occur during the tailwind and headwind part of the mission respectively.

Question 2: Sine-wave Trajectory Generation

Primarily this is done using the code in the zip folder in which this document is present and uses the dronekit library to command the drone in the guided mode. For this to be executed the Ardupilot SITL has to be launched first and the python code which guides the vehicle should be run next. The image below shows the trajectory generated.

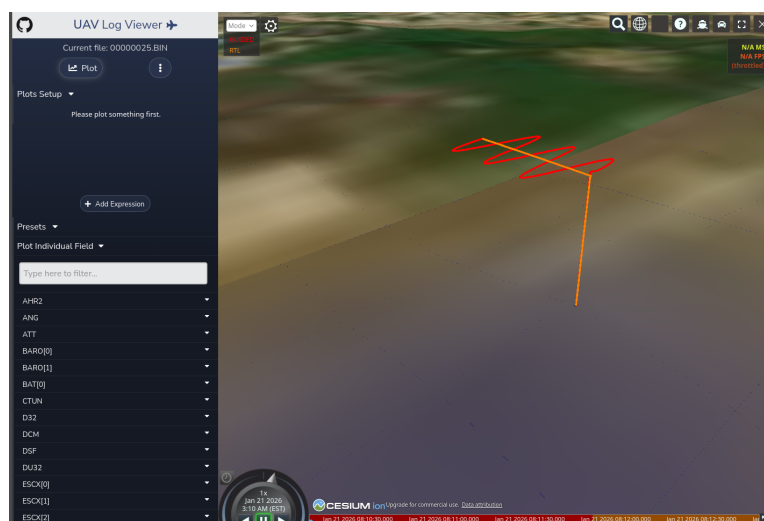


Figure 14: Complete Trajectory